

EXECUTION FLOW OF MY PROJECT SENTINEL(INCREMENTAL AUTOLOADER):-

- Simulated File Arrival: I used the data generator script to copy raw CSV files into the `raw\_landing` folder, mimicking files arriving in real-time.
- Auto Loader & Offsets: I connected the simulator to our local database, scanned the landing folder, and matched file names against our offset store. The pipeline automatically skipped already processed files to guarantee exactly-once processing.
- Raw Bronze Ingestion: I loaded the new files into memory, stripped out dummy columns caused by trailing commas, enriched the rows with metadata (file name, arrival time, ingestion time), and wrote them into the `bronze\_transactions` table.
- Dynamic Schema Evolution: I implemented schema evolution. When a file with new columns arrived, the simulator automatically altered the SQLite table schema at runtime to accommodate them without crashing.
- Silver Cleansing & Idempotent Merge: I read the Bronze data, mapped customer ID to user ID, price to amount, dropped null keys, and eliminated duplicate transaction IDs. I then executed an upsert merge—inserting new records and updating existing ones in-place.
- Gold KPI Analytics: I aggregated the clean Silver transactions to compute total records, total values, averages, freshness latency, and flagged anomalous high-value transactions. I then overwrote the Gold table with these updated KPIs.

NOTE:

1) (Cloud Orchestrations): I created and packaged the adf and databricks folders containing the template JSON configurations so that the project remains strictly bound to the original enterprise requirements for cloud runs.

2) (Free Edition Usage): Since I am using the free Databricks Community Edition and do not have a paid subscription, I run the notebooks interactively using DBFS (Databricks File System) rather than Azure Data Factory (ADF) orchestration and ADLS Gen2 storage which were specified in the corporate requirements.

OUTPUTS:

Ingesting base data:

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LOCAL DATA WAREHOUSE PREVIEW
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[Auto Loader State Store] Processed Files (1):
- Fact_Sales_1.csv (processed at: 2026-07-18 19:55:19)

[Bronze Layer] Total Raw Records: 4410
[Silver Layer] Total Cleaned/Merged Records: 4410

[Silver Layer] Top 5 Transactions Sample:
- ID: 1, User: 4, Amt: $18.29, Time: 2021-05-04 02:00:00, Method: visa
- ID: 2, User: 5, Amt: $1.49, Time: 2021-05-04 03:04:00, Method: visa
- ID: 3, User: 5, Amt: $5.89, Time: 2021-05-04 03:56:00, Method: visa
- ID: 4, User: 8, Amt: $11.59, Time: 2021-05-04 05:20:00, Method: mastercard
- ID: 5, User: 5, Amt: $12.39, Time: 2021-05-04 05:45:00, Method: mastercard

[Gold Layer] KPI Aggregate Statistics:
- Total Records Processed: 4410
- Latest Freshness Time: 2026-07-18T19:55:19.632187
- Total Transaction Value: $49984.8
- Average Transaction Size: $11.33
- High-Value Anomalies: 513
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```

Incremental merge:

```
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LOCAL DATA WAREHOUSE PREVIEW
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[Auto Loader State Store] Processed Files (2):
- Fact_Sales_1.csv (processed at: 2026-07-18 19:55:19)
- Fact_Sales_2.csv (processed at: 2026-07-18 19:57:01)

[Bronze Layer] Total Raw Records: 4450
[Silver Layer] Total Cleaned/Merged Records: 4450

[Silver Layer] Top 5 Transactions Sample:
- ID: 1, User: 4, Amt: $18.29, Time: 2021-05-04 02:00:00, Method: visa
- ID: 2, User: 5, Amt: $1.49, Time: 2021-05-04 03:04:00, Method: visa
- ID: 3, User: 5, Amt: $5.89, Time: 2021-05-04 03:56:00, Method: visa
- ID: 4, User: 8, Amt: $11.59, Time: 2021-05-04 05:20:00, Method: mastercard
- ID: 5, User: 5, Amt: $12.39, Time: 2021-05-04 05:45:00, Method: mastercard

[Gold Layer] KPI Aggregate Statistics:
- Total Records Processed: 4450
- Latest Freshness Time: 2026-07-18T19:57:01.827157
- Total Transaction Value: $50436.11
- Average Transaction Size: $11.33
- High-Value Anomalies: 517
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```

Schema Evolution test:

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LOCAL DATA WAREHOUSE PREVIEW
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[Auto Loader State Store] Processed Files (2):
- Fact_Sales_1.csv (processed at: 2026-07-18 19:55:19)
- Fact_Sales_2.csv (processed at: 2026-07-18 19:57:01)

[Bronze Layer] Total Raw Records: 4450
[Silver Layer] Total Cleaned/Merged Records: 4450

[Silver Layer] Top 5 Transactions Sample:
- ID: 1, User: 4, Amt: $18.29, Time: 2021-05-04 02:00:00, Method: visa
- ID: 2, User: 5, Amt: $1.49, Time: 2021-05-04 03:04:00, Method: visa
- ID: 3, User: 5, Amt: $5.89, Time: 2021-05-04 03:56:00, Method: visa
- ID: 4, User: 8, Amt: $11.59, Time: 2021-05-04 05:20:00, Method: mastercard
- ID: 5, User: 5, Amt: $12.39, Time: 2021-05-04 05:45:00, Method: mastercard

[Gold Layer] KPI Aggregate Statistics:
- Total Records Processed: 4450
- Latest Freshness Time: 2026-07-18T19:58:54.187822
- Total Transaction Value: $50436.11
- Average Transaction Size: $11.33
- High-Value Anomalies: 517
=====
```